Zoo DAO: Decentralized Governance for Open Science Research

Zoo Labs Foundation Research
research@zoo.ngo

Zoo Labs Foundation Research Paper ZOO-GOV-2026

With contributions from the Zoo Open Science Network

February 2026

RESEARCH

Zoo DAO: Decentralized Governance for Open Science Research

Zoo Labs Foundation Research Paper ZOO-GOV-2026

Zoo Labs Foundation Research
research@zoo.ngo

With contributions from the Zoo Open Science Network

February 2026

Abstract

Traditional scientific funding suffers from well-documented pathologies: conservative grant allocation favoring established researchers, opaque peer review processes, multi-year funding cycles misaligned with research timelines, and geographic concentration in wealthy nations. We present the **Zoo DAO**, a decentralized autonomous organization for governing open science research funding, peer review, and intellectual property. Zoo DAO employs quadratic voting for research proposal selection, sybil-resistant reputation tokens for researcher credentialing, retroactive public goods funding for high-impact discoveries, and incentivized peer review through reviewer staking. We formalize the governance mechanism, analyze its game-theoretic properties, and present simulation results from 18 months of operation on the Zoo Network. The DAO has funded 143 research projects across 34 countries, with reviewer response times averaging 8.4 days (compared to 3–6 months for traditional journals) and a researcher satisfaction score of 4.3/5.0. We demonstrate that quadratic voting allocates funds more efficiently than 1-token-1-vote systems, with a 41% higher correlation between funded projects and subsequent citation impact. All governance parameters are managed through Zoo Improvement Proposals (ZIPs) at `zips.zoo.ngo`.

Keywords: decentralized governance, quadratic voting, research funding, peer review, reputation systems, open science, DAO

Contents

1 Introduction

Scientific research funding is one of the highest-leverage activities in human civilization, yet its governance mechanisms have not fundamentally changed since the establishment of the National Science Foundation in 1950 [Bush, 1945]. The system exhibits several well-documented failure modes:

(i) **Conservatism bias**: Grant panels favor incremental research over transformative ideas, as novel proposals lack the preliminary data that reviewers demand [Boudreau et al., 2016].

(ii) **Geographic concentration**: 80% of global R&D spending occurs in 10 countries [UNESCO, 2021], excluding researchers in developing nations from funding opportunities.

(iii) **Slow cycles**: Grant application-to-funding timelines of 6–18 months are misaligned with fast-moving research areas [Alberts et al., 2014].

(iv) **Peer review opacity**: Anonymous peer review enables arbitrary gatekeeping and bias [Lee et al., 2013].

(v) **Misaligned incentives**: Researchers are rewarded for publication quantity rather than knowledge quality, producing a replication crisis [Ioannidis, 2005].

(vi) **IP monopolization**: Publicly funded research generates proprietary patents, limiting knowledge diffusion [Heller & Eisenberg, 1998].

1.1 Decentralized Science (DeSci)

The DeSci movement applies blockchain and decentralized governance principles to scientific research [Hamburg, 2019]. Key tenets include open access to research outputs, transparent funding allocation, community governance of research priorities, and token-incentivized contribution. Projects like VitaDAO (longevity research), LabDAO (computational biology), and Molecule (IP-NFTs for drug discovery) have demonstrated the viability of decentralized research coordination.

Zoo Labs Foundation extends DeSci principles to conservation science, environmental research, and AI for biodiversity—domains that particularly benefit from global coordination and community governance.

1.2 Contributions

1. A **complete governance framework** for decentralized research funding, encompassing proposal submission, review, voting, funding, and accountability.

2. **Quadratic voting** adapted for research grant allocation, with formal analysis showing superior preference aggregation.

3. A **sybil-resistant reputation system** that credentials researchers based on verified contributions.

4. **Incentivized peer review** through reviewer staking, where reviewers commit tokens to their assessments.

5. **Retroactive public goods funding** that rewards high-impact research after outcomes are demonstrated.

6. **Empirical evaluation** from 18 months of Zoo DAO operation comparing outcomes to traditional funding mechanisms.

2 Related Work

2.1 DAO Governance

DAOs coordinate collective decision-making through smart contracts. MakerDAO pioneered DeFi governance [MakerDAO, 2017]; Compound's Governor framework standardized on-chain proposal-vote-execute patterns [Compound, 2020]. However, governance participation remains low and plutocratic [Barbereau et al., 2022].

2.2 Quadratic Mechanisms

Quadratic voting (QV) [Lalley & Weyl, 2018] makes the cost of expressing preference intensity scale quadratically: K votes cost K^2 credits. This produces welfare-optimal outcomes under standard assumptions [Posner & Weyl, 2018]. Quadratic funding (QF) [Buterin et al., 2019] extends QV to public goods, where matching funds amplify small contributions. Gitcoin has deployed QF for Ethereum ecosystem grants, distributing over $50M [Gitcoin, 2023].

2.3 Reputation Systems

On-chain reputation is implemented through soulbound tokens (SBTs) [Weyl et al., 2022]—non-transferable tokens representing credentials. SourceCred [SourceCred, 2020] and Coordinape [Coordinape, 2021] provide peer attestation-based alternatives.

2.4 Science DAOs

Existing science DAOs include VitaDAO (longevity), LabDAO (computational biology), Molecule (drug discovery IP-NFTs), and ResearchHub (open science discussion). Zoo DAO differs by focusing on conservation science, incorporating multi-stakeholder governance, and implementing quadratic mechanisms designed for research allocation.

3 Governance Architecture

3.1 Token Structure

Zoo DAO uses a dual-token system:

1. **ZOO token** (fungible, transferable): Governance token for voting credits, staking, and economic coordination. Maximum supply: 1 billion.
2. **ZooRep** (non-transferable, soulbound): Reputation tokens earned through verified contributions. Determines eligibility for governance roles and review assignments.

Table 1: ZooRep earning rates by contribution type.

Contribution	ZooRep earned	Decay rate
Published research (open access)	100–500	10%/year
Peer review completion	20–50	15%/year
Data contribution (verified)	10–100	10%/year
Successful grant project	200–1000	5%/year
Citizen science observations	1–5	20%/year
Code contribution (merged)	10–50	15%/year

ZooRep earning mechanisms. Decay ensures reputation reflects recent activity.

3.2 Governance Bodies

1. **General Assembly**: All ZOO holders. Votes on constitutional changes and treasury allocation ceilings. 10% quorum.
2. **Research Council**: Members with $\geq$500 ZooRep. Reviews grants and sets research priorities. 7-member elected body, staggered 2-year terms.
3. **Review Board**: Members with $\geq$200 ZooRep in relevant domains. Conducts peer review.
4. **Treasury Committee**: 5 multi-sig holders (3-of-5) managing DAO treasury. Elected annually.
5. **Domain Working Groups**: Self-organizing groups focused on specific research areas. Minimum 5 members with $\geq$100 ZooRep.

3.3 Proposal Types

Table 2: Proposal types and governance requirements.

Type	**Proposer req.**	**Quorum**	**Voting method**	**Duration**
Research grant	50 ZooRep	5%	Quadratic	14 days
Parameter change	200 ZooRep	10%	Token-weighted	7 days
Constitutional	500 ZooRep	20%	Supermajority	21 days
Emergency	Research Council	33%	Simple majority	48 hours

4 Quadratic Voting for Research Funding

4.1 Mechanism Design

In each funding round, the DAO allocates budget B across N proposals. Voter i receives voice credits v_i proportional to ZOO holdings (capped). Voter i allocates k_{ij} votes to proposal j, paying k_{ij}^2 credits:

$$\sum_{j=1}^{N} k_{ij}^2 \leq v_i \quad \forall i \tag{1}$$

Total votes for proposal j:

$$V_j = \sum_{i=1}^{M} k_{ij} \tag{2}$$

Budget allocation proportional to squared total votes:

$$B_j = B \cdot \frac{\max(V_j, 0)^2}{\sum_{j'} \max(V_{j'}, 0)^2} \tag{3}$$

Negative votes are permitted to signal opposition.

4.2 Sybil Resistance

We implement three defenses: ZooRep gating ($\geq$50 ZooRep to vote), pairwise coordination detection [Buterin et al., 2019], and connection-weighted QV that scales credits by social graph uniqueness.

4.3 Formal Properties

Theorem 1 (Efficiency of QV for Research Allocation)**.** *Under quasi-linear utility over research outcomes and sufficient voice credits, QV funding allocation converges to the utilitarian welfare-maximizing allocation as voter count increases.*

Proof. Under quasi-linear utility $u_i(B_j) = \theta_{ij} \cdot B_j - k_{ij}^2$ where θ_{ij} is voter i's marginal valuation, the first-order condition gives $k_{ij}^* = \theta_{ij}/2$. The aggregate vote $V_j = \sum_i \theta_{ij}/2$ is proportional to total social valuation. The QF rule ensures funding proportional to the square of aggregate valuation, approximating the utilitarian optimum [Lalley & Weyl, 2018]. □

4.4 Comparison with Alternatives

Table 3: Comparison of voting mechanisms for research funding.

Property	1T1V	QV	Conviction	Expert panel
Plutocracy resistance	Low	High	Medium	N/A
Preference intensity	No	Yes	Partial	Yes
Sybil resistance	High	Medium	High	High
Participation cost	Low	Medium	Low	High
Minority voice	Low	High	Medium	Low
Scalability	High	High	High	Low

5 Incentivized Peer Review

5.1 Review Assignment

Reviewers are assigned based on domain expertise (ZooRep in relevant groups), conflict of interest checks (no co-authorship within 3 years), workload balancing (maximum 3 concurrent reviews), and geographic diversity (at least one reviewer from a different continent).

5.2 Reviewer Staking

Reviewers stake ZOO tokens to signal commitment:

$$\text{Stake}_r = \min(s_{\text{base}} + s_{\text{rep}} \cdot \text{ZooRep}_r,\ s_{\max}) \tag{4}$$

where $s_{\text{base}} = 50$ ZOO, $s_{\text{rep}} = 0.1$ ZOO per ZooRep, and $s_{\max} = 500$ ZOO.

5.3 Review Quality Assessment

Quality is assessed through author feedback (helpfulness rating), inter-reviewer consistency, and post-publication calibration (retracted positives or high-impact negatives adjust scores).

5.4 Rewards and Penalties

$$\text{Reward}_r = \begin{cases} \text{Stake}_r \times (1 + 0.15 \times Q_r) & \text{if } Q_r \geq 0.6 \\ \text{Stake}_r \times (0.5 + 0.5 \times Q_r) & \text{if } 0.3 \leq Q_r < 0.6 \\ \text{Stake}_r \times 0.3 & \text{if } Q_r < 0.3 \end{cases} \tag{5}$$

where $Q_r \in [0, 1]$ is composite quality. High-quality reviews earn up to 15% return; low-quality reviews lose up to 70% of stake.

6 Retroactive Public Goods Funding

6.1 Motivation

Prospective funding rewards proposals; retroactive funding rewards results. This reduces the risk of funding bad proposals and increases the reward for genuine public goods [Boudreau et al., 2016].

6.2 Mechanism

Quarterly, the DAO allocates R_{pool} (20% of budget) for retroactive funding. Eligible outputs: open-source software used by $\geq$5 projects, datasets with $\geq$100 uses, papers with $\geq$20 citations or verified real-world impact, and infrastructure with $\geq$6 months uptime.

Allocation uses QV with a 21-member badge holder committee ($\geq$1000 ZooRep), following the Optimism retroPGF model [Optimism, 2023] adapted for scientific impact.

6.3 Impact Metrics

$$I = w_1 \cdot \text{Reuse} + w_2 \cdot \text{Citation} + w_3 \cdot \text{Conservation} + w_4 \cdot \text{Community} \tag{6}$$

Weights set by the Research Council and updated through ZIPs.

7 Empirical Results

7.1 Operational Summary

Table 4: Zoo DAO operational statistics (18-month period).

Metric	**Value**
Total proposals submitted	287
Proposals funded	143 (49.8%)
Countries represented	34
Total funding distributed	\$4.7M equivalent
Mean project size	\$32,900
Unique voters (per round)	847 $\pm$ 214
Mean review turnaround	8.4 days
Researcher satisfaction	4.3/5.0
Published outputs	89 papers, 34 datasets, 21 tools

7.2 Funding Allocation Quality

Table 5: Funding allocation quality: QV vs. simulated 1T1V.

Metric	**QV**	**1T1V (simulated)**
Correlation with citation impact	0.67	0.47
Gini coefficient of funding	0.34	0.71
Projects from Global South (%)	38%	12%
Novel research topics funded (%)	44%	27%
Post-funding completion rate	87%	82%

QV allocation shows 41% higher correlation with citation impact, lower Gini coefficient, triple the Global South funding rate, and more novel topics funded.

7.3 Peer Review Performance

Table 6: Peer review comparison.

Metric	Zoo DAO	Traditional
Median review time	8.4 days	97 days
Review completion rate	94%	68%
Author satisfaction	4.3/5.0	3.1/5.0
Review length (words)	1,247	892
Actionable suggestions per review	4.8	2.1

Staking produces 10x faster, more complete, and higher-quality reviews.

7.4 Retroactive Funding Impact

After three rounds, 47 public goods received awards:

- 82% of funded tools saw increased adoption post-award.
- 67% of funded datasets received new citations within 3 months.
- Recipients were 2.4x more likely to contribute subsequent public goods.

8 Game-Theoretic Analysis

8.1 Voter Strategic Behavior

Under QV, strategic voting is bounded. Misrepresentation cost scales quadratically:

$$\text{Cost of misrepresentation} = \sum_j (k_{ij}^* - k_{ij})^2 \tag{7}$$

This makes strategic behavior increasingly expensive relative to sincere voting as proposals increase.

8.2 Collusion Resistance

We implement pairwise penalization:

$$V_j^{\text{adjusted}} = \sum_i k_{ij} \cdot \left(1 - \frac{1}{M} \sum_{i' \neq i} \text{sim}(i, i')^2\right) \tag{8}$$

where $\text{sim}(i, i')$ measures voting pattern overlap.

8.3 Reviewer Honesty

Proposition 1 (Honest Review Equilibrium)**.** *Under the staking mechanism, honest reviewing is a Nash equilibrium when expected quality from honest review exceeds that from strategic review by more than the effort cost difference.*

This holds because inter-reviewer consistency rewards agreement with independent (likely honest) assessors, author feedback rewards genuine engagement, and post-publication calibration penalizes divergence from realized outcomes.

9 Discussion

9.1 Lessons Learned

Participation requires simplicity. A mobile swipe-to-vote interface increased participation 3.2x over the initial web-only version.

Reputation decay is essential. Without decay, early contributors accumulate permanent governance power. Annual 10–20% decay ensures active contributor influence.

Review staking works. Staking improved quality and speed, but stake sizes require careful calibration.

Geographic diversity requires active measures. Despite QV's properties, organic participation skewed toward wealthy nations. Targeted outreach and regional ambassadors were necessary.

9.2 Limitations

1. **Scale**: 847 active voters is small relative to national agencies. Governance quality at larger scale is untested.
2. **Token volatility**: ZOO price fluctuations create budget uncertainty. Stablecoin treasury management mitigates this.
3. **Expertise concentration**: Specialized areas may lack sufficient qualified reviewers.
4. **Regulatory status**: DAO-funded research IP rights and liability vary across jurisdictions.
5. **Centralization risk**: Research Council and Treasury Committee hold significant power despite decentralized design.

10 Future Work

1. **Cross-DAO collaboration**: Inter-DAO funding protocols with VitaDAO, LabDAO, and other science DAOs.
2. **AI-assisted review**: Language models assisting reviewers with literature checks and statistical validation.
3. **Prediction markets**: Adding prediction markets for research outcomes to improve funding allocation.
4. **Decentralized preprints**: On-chain preprint server with timestamped priority claims.
5. **Research NFTs**: Non-transferable attestation of research contributions as on-chain credentials.
6. **Federated governance**: Multi-DAO governance for shared research infrastructure.

11 Conclusion

Zoo DAO demonstrates that decentralized governance can produce faster, fairer, and more impactful research funding. Quadratic voting allocates funds more efficiently by capturing preference intensity while resisting plutocracy. Incentivized peer review produces higher-quality, faster reviews. Retroactive funding rewards genuine impact over proposal writing skill.

After 18 months, the DAO has funded 143 projects across 34 countries with demonstrably better allocation quality than simulated alternatives. The Zoo DAO model provides a viable blueprint for open, community-governed science. Through Zoo Improvement Proposals at `zips.zoo.ngo`, the governance framework itself evolves through the democratic processes it enables.

References

Alberts, B., Kirschner, M.W., Tilghman, S., and Varmus, H. Rescuing US biomedical research from its systemic flaws. *PNAS*, 111(16):5773–5777, 2014.

Azoulay, P., Graff Zivin, J.S., Li, D., and Sampat, B.N. Public R&D investments and private-sector patenting. *NBER Working Paper*, 2019.

Barbereau, T., Smethurst, R., Papez, O., Lieberum, A., and Fridgen, G. DeFi, not so decentralized: The measured distribution of voting rights. *HICSS*, 2022.

Benkler, Y. *The Wealth of Networks*. Yale University Press, 2006.

Boudreau, K.J., Guinan, E.C., Lakhani, K.R., and Riedl, C. Looking across and looking beyond the knowledge frontier. *Management Science*, 62(10):2765–2783, 2016.

Bush, V. Science, the Endless Frontier. US Government Printing Office, 1945.

Buterin, V., Hitzig, Z., and Weyl, E.G. A flexible design for funding public goods. *Management Science*, 65(11):5171–5187, 2019.

Buterin, V. Ethereum: A next-generation smart contract platform. Ethereum Whitepaper, 2014.

Compound. Compound Governance. `https://compound.finance/governance`, 2020.

Coordinape. A framework for decentralized coordination. Technical report, 2021.

Franzoni, C. and Sauermann, H. Crowd science: Organization of research in open collaborative projects. *Research Policy*, 43(1):1–20, 2014.

Frey, B.S. and Gallus, J. Towards an economics of awards. *Journal of Economic Surveys*, 31(1):190–200, 2017.

Gitcoin. Gitcoin Grants: Quadratic funding for public goods. `https://gitcoin.co/grants`, 2023.

Hamburg, S. Decentralized science: Toward a new model. *Science*, 365(6450):220–221, 2019.

Heller, M.A. and Eisenberg, R.S. Can patents deter innovation? *Science*, 280(5364):698–701, 1998.

Ioannidis, J.P.A. Why most published research findings are false. *PLOS Medicine*, 2(8):e124, 2005.

Lalley, S.P. and Weyl, E.G. Quadratic voting: How mechanism design can radicalize democracy. *AEA Papers and Proceedings*, 108:33–37, 2018.

Lee, C.J., Sugimoto, C.R., Zhang, G., and Cronin, B. Bias in peer review. *JASIST*, 64(1):2–17, 2013.

MakerDAO. The Maker Protocol: Multi-collateral Dai. Technical report, 2017.

Murray, F., Stern, S., Campbell, G., and MacCormack, A. Grand innovation prize: Incentivizing solutions. *Science*, 352(6282):151–152, 2016.

Optimism Collective. Retroactive Public Goods Funding Round 3. `https://optimism.io/retropgf`, 2023.

Ostrom, E. *Governing the Commons*. Cambridge University Press, 1990.

Posner, E.A. and Weyl, E.G. *Radical Markets*. Princeton University Press, 2018.

SourceCred. A reputation protocol for open communities. `https://sourcecred.io`, 2020.

Stephan, P. *How Economics Shapes Science*. Harvard University Press, 2012.

UNESCO. UNESCO Science Report 2021. UNESCO Publishing, 2021.

Weyl, E.G., Ohlhaver, P., and Buterin, V. Decentralized society: Finding web3's soul. *SSRN*, 2022.

Wright, A. The rise of decentralized autonomous organizations. *Stanford J. Blockchain Law & Policy*, 4(2):152–237, 2021.